

VOICEBUS – SMART MOBILITY SUPPORT SYSTEM

INTRODUCTION

Public transportation is an essential service that enables people to travel conveniently and economically. However, visually impaired individuals often face significant challenges while using public buses due to the lack of accessible real-time information regarding bus routes, arrivals, and destination stops. They frequently depend on assistance from others, reducing their independence and confidence during travel.

The VoiceBus project aims to address these challenges by providing a smart mobility support system using hotspot connectivity. The system allows users to receive real-time bus information through a dedicated mobile application connected to a Wi-Fi hotspot installed inside the bus. The information is delivered through audio announcements and notifications, making it accessible for visually impaired passengers. The project focuses on creating a low-cost, scalable, and internet-independent solution that improves accessibility and safety in public transportation.

PROJECT DESCRIPTION

VoiceBus is a smart transportation assistance system designed to provide real-time bus information to visually impaired users through hotspot-based communication.

The system consists of:

- A Wi-Fi hotspot installed inside the bus.
- A web-based dashboard for managing bus information.
- An Android mobile application for passengers.
- Text-to-Speech (TTS) technology for voice announcements.

When a passenger's smartphone detects and connects to the bus hotspot, the application retrieves information such as:

- Bus number
- Route details
- Current location
- Upcoming stops
- Destination information

The information is then announced through voice notifications, helping visually impaired users travel independently without relying on visual displays or external assistance.

OBJECTIVES

The primary objectives of the VoiceBus project are:

1. To assist visually impaired individuals in using public transportation independently.
2. To provide real-time bus route and stop information through hotspot connectivity.
3. To eliminate the dependency on internet connectivity for obtaining travel information.
4. To develop a user-friendly Android application with voice assistance.
5. To create a centralized dashboard for managing bus and route information.
6. To improve accessibility and inclusivity in public transportation systems.
7. To develop a cost-effective and scalable solution suitable for widespread deployment.

METHODOLOGY

The project follows the following methodology:

Phase 1: Requirement Analysis

- Identification of challenges faced by visually impaired passengers.
- Study of existing transportation assistance systems.
- Analysis of hotspot-based communication methods.

Phase 2: System Design

- Designing the overall architecture.
- Planning communication between hotspot, dashboard, and mobile application.
- Designing user interfaces for both web dashboard and Android application.

Phase 3: Web Dashboard Development

- Creation of a responsive web dashboard.
- Display of bus details, routes, and upcoming stops.
- Simulation of real-time bus information.

Phase 4: Android Application Development

- Development of Android application using Android Studio.
- Implementation of hotspot detection.
- Integration of Text-to-Speech functionality.
- Creation of accessible user interfaces.

Phase 5: Testing and Validation

- Verification of hotspot connectivity.
- Testing of audio announcements.
- Validation of route and stop information display.

WORK COMPLETED

The following tasks have been completed successfully:

Literature Study

- Studied accessibility challenges faced by visually impaired users.
- Analyzed existing smart transportation systems.

Dashboard Development

A web-based Smart Bus Information Dashboard has been developed featuring:

- Smart Bus Information System interface.
- Hotspot connection status indicator.
- Bus number display.
- Route information display.
- Upcoming stop timeline.
- Modern high-contrast user interface.

System Architecture Design

- Designed communication flow between bus hotspot and user application.
- Planned real-time information exchange mechanism.

Android Studio Setup

- Android development environment configured.
- Project structure created.
- Initial screens and layouts prepared.

CURRENT PROGRESS

The VoiceBus project is currently in the final stages of software development and integration. Significant progress has been achieved in both the web dashboard and Android application development.

Completed Tasks

- ✓ Problem analysis and requirement gathering
- ✓ System architecture design
- ✓ User interface design for accessibility
- ✓ Development of Smart Bus Information Web Dashboard
- ✓ Implementation of hotspot connectivity concept
- ✓ Bus route and upcoming stop information display
- ✓ Android Studio project setup and configuration
- ✓ Android application user interface development
- ✓ Text-to-Speech (TTS) integration and testing
- ✓ Voice announcement functionality for bus and stop information
- ✓ Testing of dashboard functionalities

Ongoing Tasks

- 🔄 Android application feature refinement
- 🔄 Hotspot communication testing between bus module and mobile application
- 🔄 User experience improvements for visually impaired passengers

Overall Project Status

The project has achieved approximately **90% completion**. The core functionalities, including the web dashboard, hotspot-based communication concept, and Text-to-Speech voice assistance, have been successfully developed and tested. Current efforts are focused on final integration, optimization, and preparing the system for demonstration and deployment.

CHALLENGES FACED

During development, several challenges were encountered:

1. Designing an accessible interface suitable for visually impaired users.
2. Establishing reliable hotspot communication between bus and mobile devices.
3. Integrating real-time updates without internet dependency.
4. Managing Android Studio configurations and Gradle dependencies.
5. Ensuring smooth Text-to-Speech functionality across different Android devices.
6. Synchronizing dashboard information with mobile application data.

SOLUTIONS IMPLEMENTED

The following solutions were adopted:

1. High-contrast UI design was implemented for better visibility.
2. Hotspot-based communication was selected to avoid internet dependency.
3. Android Text-to-Speech API was integrated for voice announcements.
4. Modular software architecture was adopted for easier maintenance.
5. Responsive web technologies were used for dashboard development.
6. Real-time update mechanisms were designed for future scalability.

RESULTS

The developed system has successfully demonstrated:

Web Dashboard

- Display of bus number.
- Route visualization.
- Upcoming stop tracking.
- Hotspot connection monitoring.

Expected Android Application Features

- Automatic hotspot detection.
- Voice-based bus announcements.
- Stop alerts.
- Route information retrieval.

Benefits

- Increased independence for visually impaired passengers.
- Reduced dependence on external assistance.
- Low-cost implementation.
- Offline operation capability.

FUTURE WORK

The project can be enhanced through:

1. Integration of GPS-based live bus tracking.
2. AI-based route prediction and travel assistance.
3. Multi-language voice announcements.
4. Cloud-based monitoring system.
5. Emergency alert functionality.

6. Support for hearing-impaired passengers through vibration notifications.
7. Integration with government transportation databases.
8. Deployment across multiple buses and routes.

CONCLUSION

VoiceBus is a smart mobility assistance system designed to improve accessibility in public transportation for visually impaired individuals. By utilizing hotspot-based communication and voice notifications, the system provides real-time travel information without requiring continuous internet connectivity. The developed web dashboard successfully demonstrates route and stop management capabilities, while the Android application is under development to provide complete user interaction and accessibility features. The project offers a cost-effective, scalable, and user-friendly solution that has the potential to significantly improve independent travel for visually impaired passengers.